# Impact Analysis and Attack Simulation on Quishing (a QC Code Phishing) using QRLJacker

ERIC BLANCAFLOR
*School of Information Technology*
*Mapua University*
Makati, Philippines
ebblancaflor@mapua.edu.ph

MARK BRANDON CALIDA
*School of Information Technology.*
*Mapua University*
Makati, Philippines
mbmcalida@mymail.mapua.edu.ph

MATTHEW CHAN
*School of Information Technology.*
*Mapua University*
Makati, Philippines
matthewchan@mymail.mapua.edu.ph

NICOLE LAYSICO
*School of Information Technology*
*Mapua University*
Makati, Philippines
namlaysico@mymail.mapua.edu.ph

KEITH GOTEN BON SUPAN
*School of Information Technology*
*Mapua University*
Makati, Philippines
kgbosupan@mymail.mapua.edu.ph

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***Abstract*— Quick response (QR) codes is a form of two-dimensional matrix barcode that can store information in a grid of black and white pixels which creates a unique pattern. The QR codes are usually scanned by mobile devices such as smartphones and will redirect the user's mobile devices to whatever URL the QR code contains. QR codes have grown in popularity due to their capabilities of redirecting to digital information or authenticating users [4]. This study implemented an explanatory research design. The study aims to describe, showcase, and investigate the different methods of quishing. To supplement the study, the researchers conducted a survey to better assess the knowledge of the peers around them. A survey was conducted to understand what the population thinks about QR codes and the underlying risks on its usage. The results of the QRJacking simulation were successful; the tool gained access to the target's Whatsapp. Outside of the success of the attack simulation, the research found that there were no restrictions when sending the Malicious QR code over messaging applications like Facebook Messenger, Discord, Whatsapp, and Viber. For future researchers, the researchers recommend using different apps, tools, and frameworks. Look into countermeasures and other methods of mitigation.**

***Keywords— QR codes, Quishing, QRJacking, Phishing,***

## I. Introduction

### *A. Background of the Study*

Quick response (QR) codes is a form of two-dimensional matrix barcode that can store information in a grid of black and white pixels which creates a unique pattern. The QR codes are usually scanned by mobile devices such as smartphones and will redirect the user's mobile devices to whatever URL the QR code contains. QR codes have grown in popularity due to their capabilities of redirecting to digital information or authenticating users [1]. QR codes have been used to address users to e-commerce sites, surveys, images, videos, etc. However, QR codes can also be manipulated to divert users to a malicious site which is exploited by phishers to conduct Quishing[2]. QR Code phishing or Quishing, like most phishing types, is a type of social engineering attack wherein the attacker pretends to be from a legitimate source and sends a QR code to the target that exposes the user's confidential data when scanned.

The use of QR codes offers convenience however it also raises security concerns [3]. Certain chatting applications have adapted the use of QR codes for login authentication. This feature can be exploited by attackers to bypass the typical username and password authentication used for accounts; this type of attack is called QRLJacking [4]. With the increasing popularity and usage of QR codes, vulnerabilities are becoming more prevalent as new tools are being used to gain access to user's information. This brings a point toward the improvement of security on applications that utilize QR codes for its features.

In recent years, two-dimensional barcodes, known as QR codes, have grown in popularity as a rapid and simple method of accessing information. Numerous types of data can be included in QR codes, such as URLs, phone numbers, and product details [5]. However, due to their widespread use, hackers are now interested in finding ways to exploit any potential weaknesses they may have.

QRL jacking is a QR code phishing attack where attackers hijack the authentication process using malicious code or software to redirect users to a fake website or app. Attackers use a valid QR code to entice people to scan it in this attack. Instead of taking the user to the desired page, the redirection instead takes them to a fake website where the attackers can steal their personal data or insert malware [5].

One such vulnerability is the phenomenon of QRL jacking, where cybercriminals hijack legitimate QR codes for their malicious purposes [6]. By replacing the legitimate QR code with a malicious one, cybercriminals can redirect users to a fake website or application, from which they can steal sensitive information or distribute malware [6]. Therefore, it is essential to understand the risks associated with QR code usage and take proactive measures to prevent QR code phishing and QRL jacking [6]. This study aims to identify the various tactics used by cybercriminals to exploit QR code vulnerabilities and recommend effective strategies to mitigate these risks.

The risk of QR code phishing is prevalent in the use of mobile learning. Teachers and instructors often use QR codes to

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develop teaching materials that students can access online [7]. While this is an innovative and engaging way to teach, it also exposes learners to a significant risk of being targeted by QR code phishing attacks. Thus, it is crucial to understand the risks associated with QR code phishing and QRL jacking in mobile learning environments to protect learners from cyber threats [7]. This study aims to explore the risks associated with QR code phishing and QRL jacking in mobile learning and provide recommendations to mitigate these risks.

### *B. Research Objective*

The following items listed below are the target of this study.

- To simulate QRLJacking attacks on BYOD devices.
- To determine the QR-based login system's weaknesses and risks.
- To suggest defenses against attacks on QR-based login systems known as QRLJacking.

### *C. Scope*

The scope of the study is focused on the in-depth analysis of the tools used in conducting Quishing and QRLJacking attacks and the processes of how they are used. The study also seeks to identify possible weaknesses of chat applications that use QR-based login systems. The study does not focus on the social engineering context of how these attacks are distributed.

## II. Review of Related Literature And Studies

### *A. Social Engineering Attacks and Phishing*

Social Engineering utilizes the psychological behavior of humans to manipulate, deceive or influence victims into giving away their personal information or control over a system[8]. It exploits human error and the tendency of people to be trusting towards others[9]. Unlike computers, people can be affected by psychological manipulation which attackers can use to gain the trust of their targets and then create a scenario wherein the victim would have to break security protocols for the attacker[10]. According to recent research and polls, social engineers have a high success rate while conducting cyberattacks, accounting for 84% of all attacks[11].

Phishing is a type of social engineering attack that tricks the victim by pretending to be a trustworthy entity that asks victims for their personal information or credentials[12]. Typically these attacks create situations wherein the victim would feel a sense of urgency, curiosity, or fear in order for victims to irrationally follow what is being asked of them. It may sound easy to avoid, however the continuous advancement of phishing techniques makes it more difficult to identify from a victim's standpoint[13].

Phishing attacks have become more sophisticated over the years, making them harder to detect. In a study by Verma et al, it highlights the importance of educating individuals about the risks associated with phishing attacks and the need for organizational measures to protect against them. Additionally, the author discusses several techniques used in phishing attacks, such as pretexting, baiting, and spear-phishing [14].

### *B. Quishing*

Quishing is a type of scam similar to phishing, smishing, and vishing. It involves a cybercriminal sending you an email or text message claiming that you have won a prize or have urgent issues with your account that need to be addressed. The message will include a QR code that directs you to a website where you are asked to enter your personal data to collect your prize or resolve the issue [15]. However, this website is actually a fake version of a legitimate website, designed to collect your personal information so that the cybercriminals can use it to steal your money.

The act of "quishing" or using malicious QR codes to conduct phishing attacks has been around for some time but so far affected a small number of victims and thus never received much of an attention. However, the COVID-19 pandemic has popularized the use of QR codes as a means of sharing information, making them more vulnerable to malicious attacks [16]. Examples of quishing attacks include the use of fraudulent QR codes on parking meters in San Antonio and the attachment of malicious QR codes to emails in a phishing campaign to collect Microsoft credentials. Quishers have also created a backdoor in an open-source QR code generator that contains malware to remotely execute code on a compromised machine or access a remote shell [16].

Quishing attacks may not be easily detected due to the fact that URL shorteners make it more difficult for users to identify the nature of the URL they would be directed to by the QR code[1]. Users would have no idea if the shortened URL contained would be malicious or not since it does not show any resemblance of the original URL. On the other hand, long URLs may not be entirely displayed due to the mobile device's operating system favoring screen usability and navigation since the narrow screen size of mobile devices cannot display the entirety of the URL without clotting most of the screen[2].

It was found that phishing attacks account for a significant percentage of overall cyber-attacks and are a significant challenge to cybersecurity [14]. QR Code Authentication with Embedded Message Authentication Code proposes a solution to combat QRL phishing by using an embedded message authentication code (EMAC) in QR codes. The EMAC serves as a digital signature, verifying the authenticity of the QR code and preventing any unauthorized tampering or modifications [5].

Therefore, it is essential to understand the threat of QRL phishing and implement measures such as QR code authentication systems to prevent such attacks' success. This study aims to investigate the effectiveness of QR code authentication with EMAC in preventing QRL phishing and provide insights on how organizations can better protect their customers from this threat [5].

### *C. QRLJacking*

QRLJacking is a social engineering attack that specifically targets apps that employ "Login with QR code" as a login method [4]. The strategy behind the attack is to lure the

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victim into scanning a malicious QR code, which would then result in the hijacking of their session and give the attacker unauthorized access to their account.

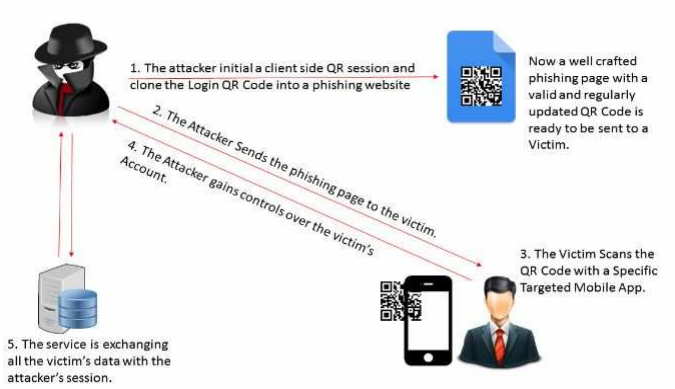


Fig. 1. QRLJacking Attack Vector [4]

The QRLJacking attack process begins with the attacker creating a clone of the Login QR Code. The created QR clone is manipulated so that the attacker would get the information instead of the real service provider. The attacker then creates a phishing page with the manipulated QR code. The phishing page would look entirely similar to the original service provider to mislead the victims into believing that it is the real page. Once the QR code is scanned by the victim, the attacker would be given access to the user's account. The attacker can then conduct different kinds of cybercrimes such as fraud and identity theft with the stolen account.

The primary reason how QRLJacking effectively works is due to the low requirements needed to perform the attack[4]. The attacker would only need to clone the expirable QR Codes and display it to the phishing webpage. The requirements would only need an attacker with a script kiddie coding level as minimum, a QR Code Refreshing Script and a phishing webpage that looks similar to a real website.

## III. METHODOLOGY

This study implemented an explanatory research design. The study aims to describe, showcase, and investigate the different methods of quishing. The researchers decided on using Metaspolit and QRLJacker to aid in conducting a simulated attack.

To supplement the study, the researchers conducted a survey to better assess the knowledge of the peers around them. Using the User Experience Questionnaire (UEQ), the researchers would be able to understand what the population thinks about QR codes. The survey includes questions about the age of the user, their preferences when it comes to QR codes, where the 30 respondents encounter them, and why they use QR codes. The respondents were gathered from convenience sampling. Convenience sampling was chosen to conform to time constraints and convenience. The survey was made using google forms and distributed online.

The researchers used the QRLJacker framework to simulate a quishing attack. Using a virtual machine, The team installed and ran the QRLJacking tool kit via GitHub.

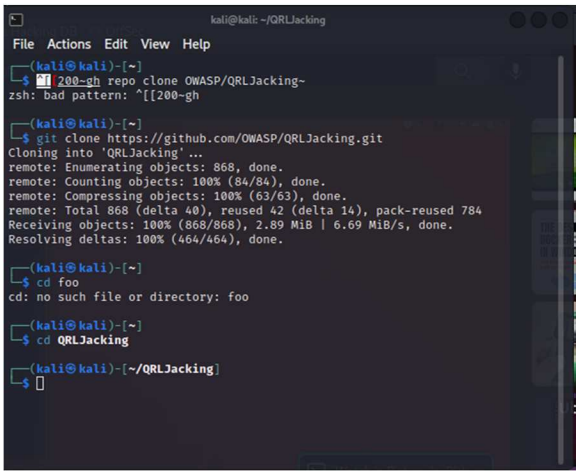


Fig. 2. Installation of QRLJacker

As presented in figure 2, it can be seen that the team is in the process of installing the toolkit in Kali Linux. For this demonstration, the researchers opted to use the Whatsapp QRLJacker tool. Here the attacker can generate a malicious QR code that can be used to log in Whatsapp. The attacker can create the host link to be LAN or WAN, the tool will still generate a QR code that grants access to the victim's chats on Whatsapp. The HTML code of the site itself is customizable, so the attacker can create a more convincing website to bait the user.

Fig. 3. Generated malicious QR code

Figure 3 shows the malicious QR code generated and a step-by-step procedure in distributing the code. Here the HTML website can be modified to appear more genuine and increase credibility to appear legitimate towards potential victims.

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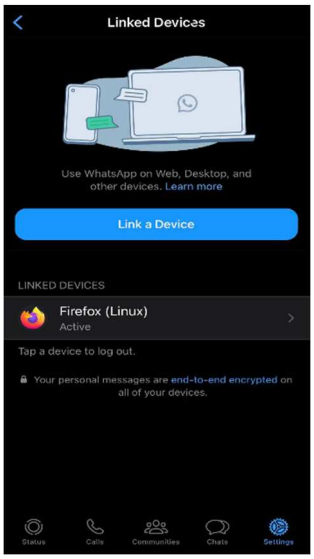


Fig. 4. Whatsapp mobile application

As presented in figure 4, a screenshot of the linked devices in a messaging app account can be seen. The attacker using Linux can see the chats and account of the victim. Not pictured here due to the account being new and did not display anything.

## IV. Results and Discussions

To know the insights of users who experienced the use of QR and its security implications, a survey was conducted in this study. The researchers garnered 100 respondents with the range of respondents' ages being 16 - 31 years old.

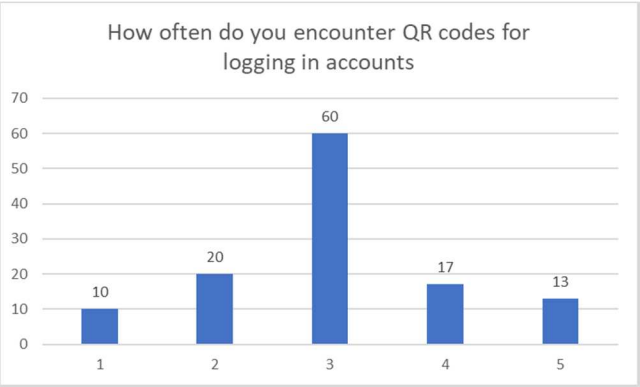


Fig. 5. How often do you encounter QR codes for logging in accounts?

Figure 5 shows how often the respondents encounter QR codes ranging from 1 to 5. 1 being never and 5 being always.

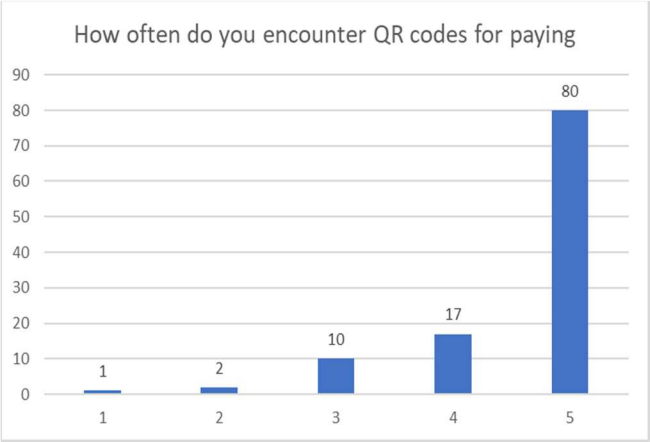


Fig. 6. How often do you encounter QR codes for paying?

Figure 6 shows how often the respondents encounter QR codes specially for paying. This question was asked with the rising trend of cashless payment options. 10% of the respondents still are yet to encounter these methods while 80% of the population encounter cashless payment options frequently.

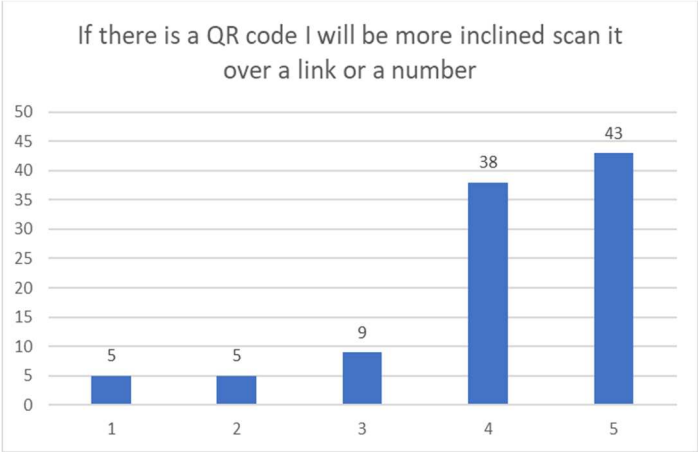


Fig. 7. If there is a QR code I will be more inclined scan it over a link/number

As shown in figure 7, the authors asks the user if they agree or disagree with the statement above. 1 being strongly disagree and 5 being strongly agree. 81% of the respondents find themselves agreeing with the sentiment above.

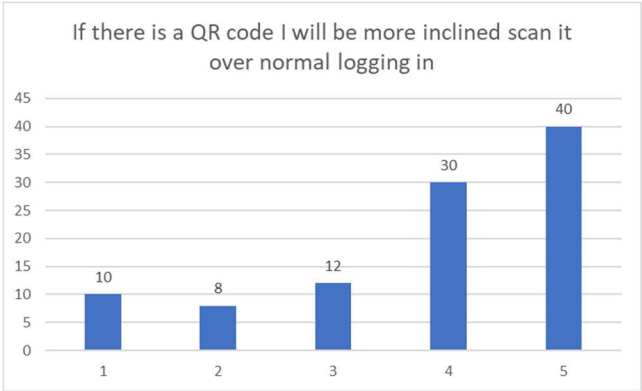


Fig. 8. I am more inclined to use QR login over normal logging in

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Figure 8 asks the user if they prefer logging in via QR code over the standard of inputting valid credentials. This result tells the researchers that the sample size is divided but the majority agree to the query

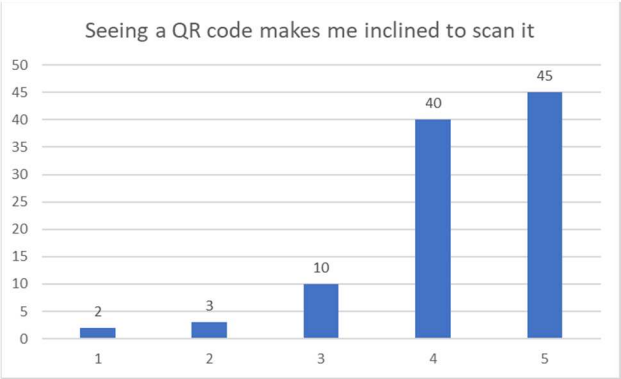


Fig. 9: Seeing a QR code makes me inclined to scan it

Figure 9 asks the user if they need to scan any QR code they might see. Like the previous result, over 85% of the sample population feels the need to check QR codes they might find in public.

## V. Conclusions and Recommendations

This study presents the QRJacking simulation, and the results were successful. The accounts the researchers used to distribute the malicious QR codes are still active and unbanned for a short period of time. The primary means of mitigating this type of social engineering attack would be to be cautious of any websites. Antivirus software that keeps track of malicious sites and updated virus definitions is also beneficial. Finding a direct way to counteract squishing is difficult, but it is best to approach quashing how you would, similar to social engineering attacks.

Quishing is one social engineering attack that should be researched more. Other similar methods have been studied in depth and have established countermeasures. Quishing is relatively new compared to social engineering attacks; thus, countermeasures are yet to be implemented and understood. QRLJacking is just one of the many tools that can be used to conduct this attack, and it not only encapsulates WhatsApp but Weibo, WeChat, and other messaging apps. For future researchers, the researchers recommend using different apps, tools, and frameworks. Look into countermeasures and other methods of mitigation.

## References


[1] T. Vidas, et al. (2013). QRishing: The Susceptibility of Smartphone Users to QR Code Phishing Attacks. Retrieved from https://link.springer.com/chapter/10.1007/978-3-642-41320-9_4

[2] F. Sharevski et al. (2022). Phishing with Malicious QR Codes. Retrieved from https://dl.acm.org/doi/abs/10.1145/3549015.3554172 (2022)

[3] G. A. Amoah et al (2022). QR Code Security: Mitigating the Issue of Quishing (QR Code Phishing). Retrieved from https://www.researchgate.net/publication/364593009

[4] SymbianSyMoh. (2016). GitHub - OWASP / QRLJacking. Retrieved from https://github.com/OWASP/QRLJacking

[5] G. Varshney, M. Misra, & P. Atrey (2018). Secure authentication scheme to thwart RT MITM, CR MITM and malicious browser extension based phishing attacks. Retrieved from https://www.sciencedirect.com/science/article/abs/pii/S2214212618300140

[6] N. Manna. (2022). Encrypted Message Traversal using QR codes. Retrieved from https://ijrpr.com/uploads/V3ISSUE10/IJRPR7472.pdf

[7] Y-J Tu, W. Zhou, & S. Piramuthu (2021). Critical risk considerations in auto-ID security: Barcode vs. RFID. Decision Support Systems. Retrieved from https://www.sciencedirect.com/science/article/abs/pii/S0167923620302268

[8] R. Kalniņš et al (2017). Security Evaluation of Wireless Network Access Points. Applied Computer Systems, 21(1), 38–45. Retrieved from https://www.researchgate.net/publication/318180928

[9] F. Salahdine, & N. Kaabouch (2019). Social Engineering Attacks: A Survey. Future Internet, 11(4). Retrieved from https://www.mdpi.com/1999-5903/11/4/89#B1-futureinternet-11-00089

[10] N. N. Pokrovskaia, & S.O. Snisarenko (2017). Social engineering and digital technologies for the security of the social capital' development. Retrieved from https://ieeexplore.ieee.org/document/8085750

[11] G. Costantino (2018). CANDY: A Social Engineering Attack to Leak Information from Infotainment System. Retrieved from https://ieeexplore.ieee.org/document/8417879

[12] M. Masoud, Y. Jaradat & A.Q. Ahmad (2016). On tackling social engineering web phishing attacks utilizing software defined networks (SDN) approach. Retrieved from https://ieeexplore.ieee.org/abstract/document/7863679

[13] I. Vayansky & S. Kumar (2018). Phishing – challenges and solutions. Computer Fraud & Security, 2018(1), 15–20. Retrieved from https://www.sciencedirect.com/science/article/abs/pii/S1361372318300071

[14] R. S. Verma, B. R. Chandavarkar,., & P. Nazareth (2019). Mitigation of hard-coded credentials related attacks using QR code and secured web service for IoT. Retrieved from https://ieeexplore.ieee.org/document/8944592

[15] Bankers Association of the Philippines. (October 2022). Quishing: A New Scam You Should Be Aware Of. Retrieved from https://bap.org.ph/cybersecurity/quishing-a-new-scam-you-should-be-aware-of/

[16] F. Sharevskiet al. (2022). Phishing with Malicious QR Codes. Retrieved from https://eurousec2022.secuso.org/pre-proceedings/eurousec2022-final22.pdf


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